ETHNOHACK 2026 · TRACK 1 · TRADITIONAL KNOWLEDGE

EthnoAnatolia Cross-Ref

From Dioscorides of Anazarbus to Anatolian folk medicine today: two thousand years of medicinal plant use, side by side

TEAM BOTANISTS

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THE QUESTION

1ST CENTURY AD · ANAZARBUS, CILICIA

Dioscorides writes *De materia medica*, the foundational text of pharmacy



2018–2020 · TÜRKIYE

TEMDB records how Anatolian people use medicinal plants today

Which of his uses are still alive in Anatolia today?

WHY IT MATTERS

Ethnobotanical knowledge is scattered across sources that do not speak the same language

For researchers

Ancient and modern records compared from one standardised, ICD-11-coded source instead of source by source

For drug discovery

A use that persisted for two millennia is a signal worth testing in the lab, not a proof

For communities

Documents Anatolian traditional knowledge and its cultural continuity before it is lost

DATA AND DISCLOSURE

What existed before, and what we built this week

PRE-EXISTING · USED UNCHANGED

- **TEMDB**, Turkish Ethnomedicinal Database, published by our team (Zenodo, 27 Aug 2026, CC BY 4.0)
- TEMDB's ICD-11 coding, completed in October 2026
- **Sprengel 1829** Latin edition (public domain) and **Wellmann 1906** Greek edition (verification only)

NEW · 2–8 NOVEMBER 2026

- **606 use records** extracted for 46 species shared by both sources
- New ICD-11 coding of every historical use, with TEMDB's rules
- Cross-reference, continuity metrics and the Colab notebook

METHOD

One coding language for two thousand years

01

Extract

Uses and preparations read from Sprengel's Latin, checked against Wellmann's Greek

02

Match taxa

Ancient plant names linked to accepted species; 46 shared with TEMDB

03

Code ICD-11

Every use gets a WHO ICD-11 MMS code under TEMDB's rules

04

Cross-ref

Same species compared at ICD-11 category, block and chapter level

05

Measure

Continuity metrics, recomputed and checked in the notebook

Shared coding rules, e.g. burns → NE2Z · wounds → ND56.1 · bites → NE61 · "purgative" → Unclassified

RESULTS · 46 SPECIES · 1,229 TEMDB RECORDS · 588 HISTORICAL USES

Half of today's uses fall in a disease chapter Dioscorides already treated

Share of TEMDB records whose ICD-11 group was also described by Dioscorides

Category	**16.0 %**
Block	**25.8 %**
Chapter	**51.9 %**

Cumulative: a category match is also a block and chapter match

19.9 %

of Dioscorides' therapeutic uses are still recorded in Anatolian folk medicine today

6 · 18 adverse-effect records excluded from the 606, giving 588 therapeutic uses

RESULTS · PER SPECIES

95 plant–disease pairs link the two sources

Highest continuity: TEMDB records in a category Dioscorides described

Hyoscyamus niger	**75 %**
Artemisia absinthium	**62.5 %**
Ficus carica	**56 %**
Punica granatum	**55.6 %**
Plantago major	**44.4 %**

8 of 46 species share no ICD-11 category at all

ONE SHARED PAIR

Achillea millefolium

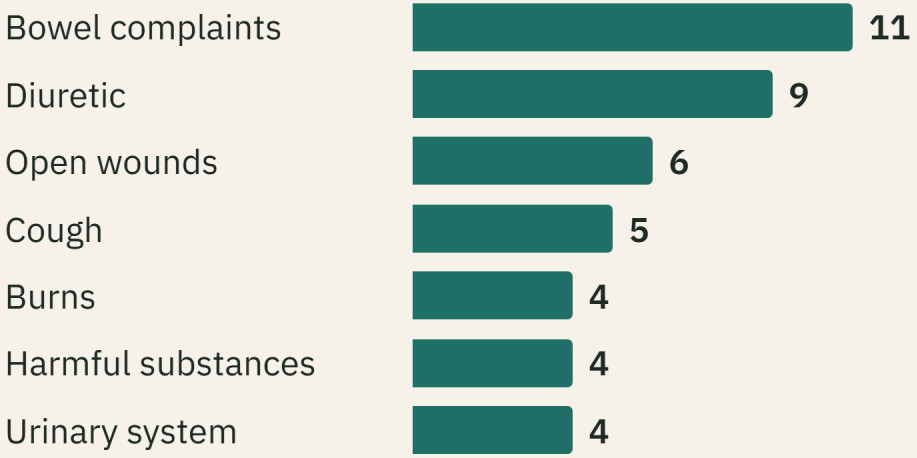
Dioscorides: bleeding wounds; prevents wound inflammation

TEMDB today: wound healing, 4 records

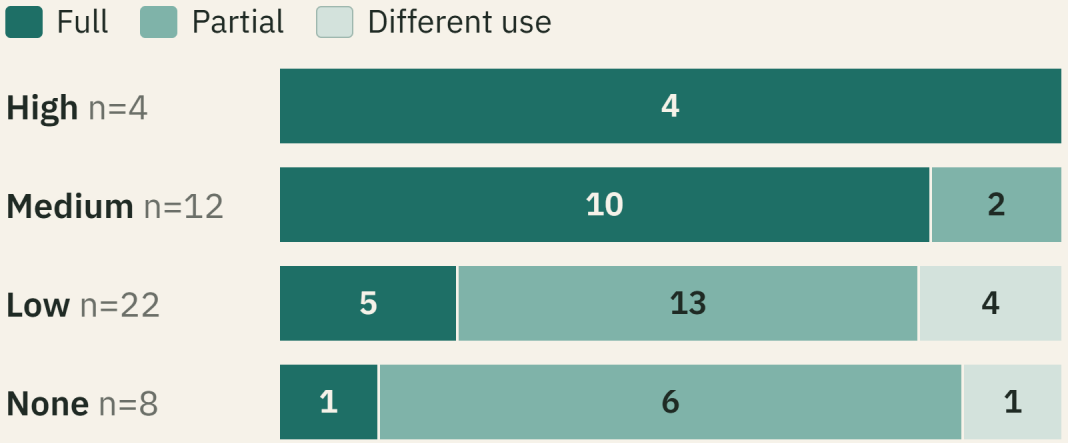
ICD-11 ND56.1 · Open wound

What lasted, and does the metric agree?

Species in which Dioscorides and TEMDB share the ICD-11 category



Expert continuity status, by objective threshold class



DEMO · IMPACT · FUTURE DEVELOPMENT

What it does today, and where it goes next

[icon: Play]

Demo

- Open in Colab, Run all
- Data checks, metrics, cross-reference table and charts rebuilt from the raw workbook
- Results downloaded as a formatted Excel file

Video: youtu.be/baJch1sK8lA

[icon: Target]

Potential impact

- 95 plant–disease pairs as a shortlist for pharmacological testing
- A reusable method: any historical text versus any ICD-11-coded modern dataset
- Anatolian traditional knowledge documented with its 2,000-year history

[icon: ArrowRight]

Future development

- Independent expert review of extractions and ICD-11 codes
- All of *De materia medica* and more historical texts
- Link species to compounds via the hackathon APIs (WFO, COCONUT, ChEBI)

LIMITATIONS AND RESPONSIBLE USE

What these numbers do not say

Absence is not loss

TEMDB covers 2018–2020 literature only; a missing use may simply be outside that window

Ancient names are uncertain

3 of 46 identifications are disputed and flagged; an ancient disease name is not a modern diagnosis



Historical claims are not medical advice

Continuity shows what people have said about a plant, not that it is safe or effective. No doses, no recipes.